MSE 5760 — Homework 5

Classification of Ti-6Al-4V Microstructures using CNN, MLP, PCA + MLP/SVM/RF Models

Spring 2026

Abstract

We implement and compare seven classification pipelines — a baseline CNN, an improved CNN, a deep MLP, PCA(100)+MLP/SVM/RF, and PCA(2)+SVM/RF — on a Ti-6%Al-4%V microstructure dataset (1225 images, 3 classes: *acicular*, *bi-modal*, *lamellae*). Baseline CNN reaches 84.08% accuracy on the 245-image test set but completely fails on the minority *acicular* class. Adding a third convolutional layer with dropout raises accuracy to 99.18%, outperforming the $\sim 93\%$ reported in the reference paper. In Part 2, classical models that use PCA features of raw pixels max out around 83–90%, while switching to HOG (Histogram of Oriented Gradients) features pushes both SVM and RF above 98%, matching the improved CNN.

Dataset and Preprocessing

The dataset (`microstructure_dataset`) contains 1225 SEM micrographs distributed across three heat-treatment classes: *acicular* (186), *bi-modal* (351) and *lamellae* (688). All images are resized to 128×128, giving an input tensor of shape `(1225, 128, 128, 3)` and label array of shape `(1225,)`. A stratified 80/20 split yields **980 training** and **245 test** images.

Part 1 — CNN Model

1.1 Load dataset (resize to 128×128) and print tensor / label shapes

After loading all 1225 images and labels we obtain `images.shape = (1225, 128, 128, 3)` and `labels.shape = (1225,)`. After the stratified 80/20 split, the input tensor of one training batch has shape `[200, 3, 128, 128]` and the ground-truth label batch has shape `[200]`.

1.2 Train/test split, raw samples, intensity histograms and preprocessing

Figure 1 shows one raw training image per class. The raw training set statistics are $\min = 1,\ \max = 255,\ \mu = 89.61,\ \sigma = 40.11$. After Z-score normalisation followed by min-max rescaling to $[0, 1]$, the training set statistics become $\min = 0.0,\ \max = 1.0,\ \mu = 0.349,\ \sigma = 0.158$. Pixel-intensity histograms before and after preprocessing are shown in Figure 2, and preprocessed samples in Figure 3.

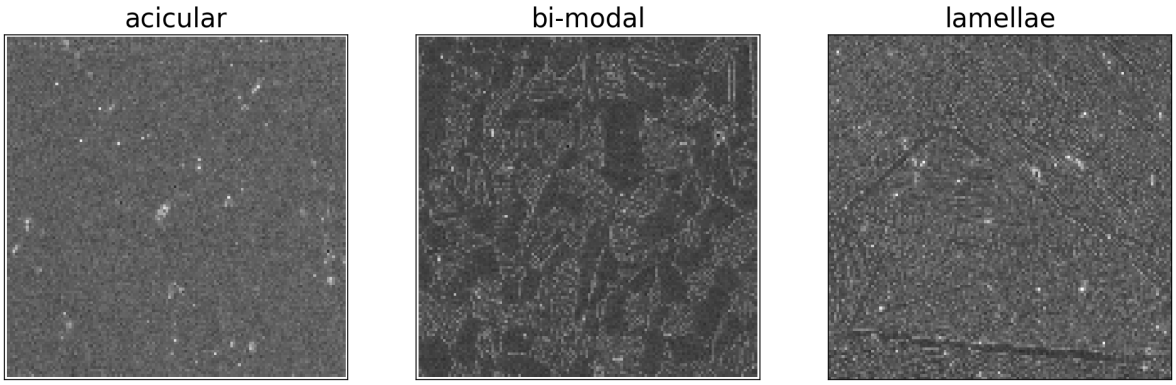
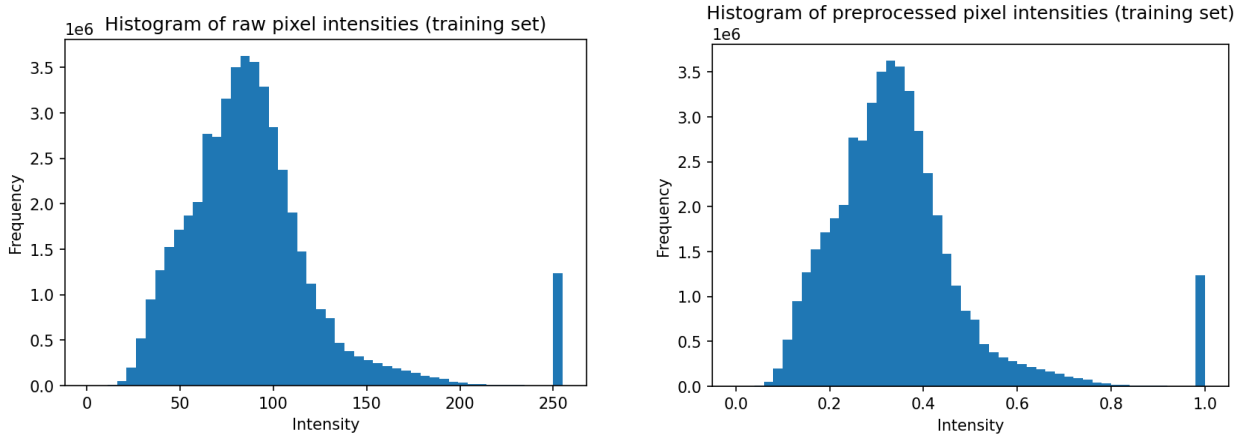


Figure 1: Raw training samples, one per class.



(a) Raw intensities.

(b) Preprocessed intensities ($[0, 1]$).

Figure 2: Pixel-intensity histograms of the training set before and after preprocessing.

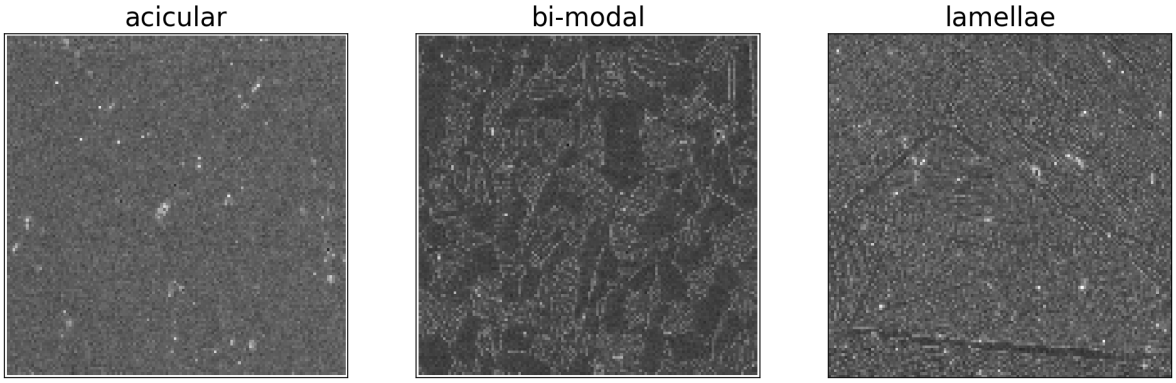


Figure 3: Preprocessed training samples, one per class.

1.3 CNN architecture and training/test loss

The CNN has two convolutional blocks followed by a fully connected layer:

- **Conv1**: $3 \to 16$ channels, 3×3 kernel, stride 1, padding 1, ReLU, MaxPool(2).
- **Conv2**: $16 \to 32$ channels, 3×3 kernel, stride 1, padding 1, ReLU, MaxPool(2).
- **FC**: $32 \cdot 32 \cdot 32 = 32{,}768 \to 3$, softmax.

Optimiser: Adam, lr $= 10^{-3}$; loss: cross-entropy; batch size 200; 100 epochs. The training/test loss is shown in Figure 4.

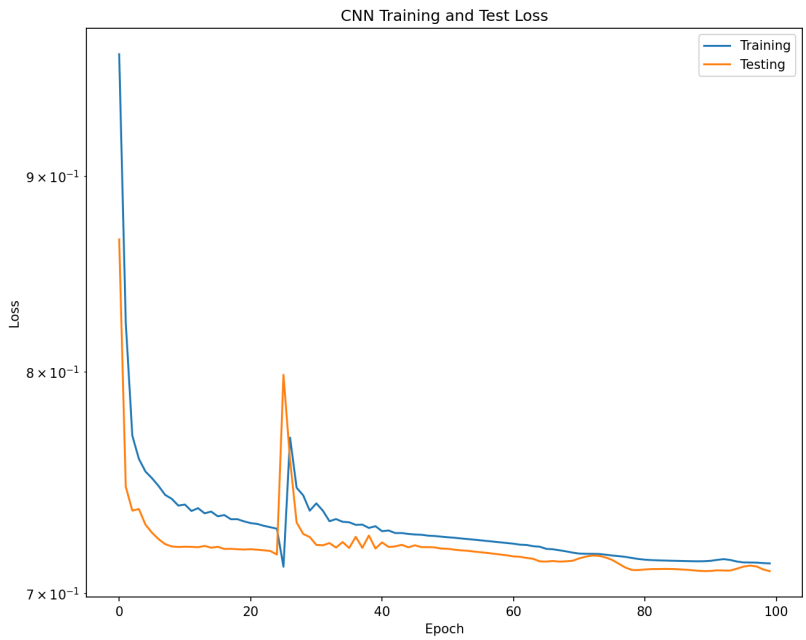


Figure 4: CNN training and test loss (log scale) over 100 epochs.

1.4 Confusion matrix, performance scores, sample predictions

On the 245-image test set, the CNN classifies 206 images correctly, giving an overall accuracy of **84.08**%. Per-class precision, recall and F1 are summarised in Table 1, and the confusion matrix in Figure 5. Correctly and incorrectly classified examples are shown in Figures 6–7.

Table 1: Classification report for the baseline CNN (test accuracy = 0.8408).

Class	Precision	Recall	F1-score	Support
acicular	0.00	0.00	0.00	37
bi-modal	0.73	0.97	0.83	70
lamellae	0.91	1.00	0.95	138
macro avg	0.55	0.66	0.60	245
weighted avg	0.72	0.84	0.77	245

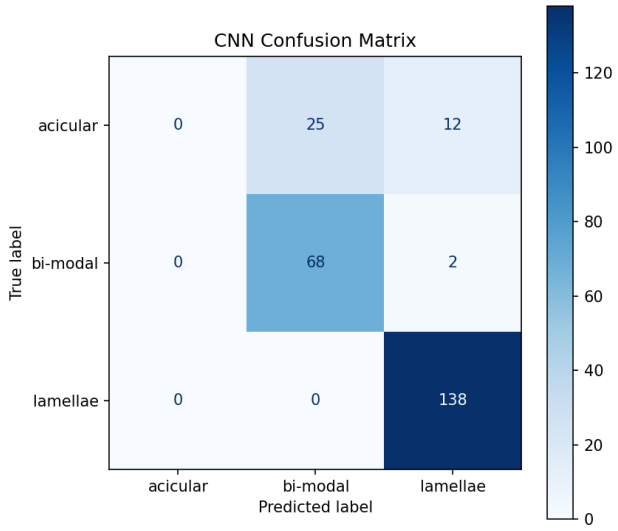


Figure 5: Confusion matrix for the baseline CNN.

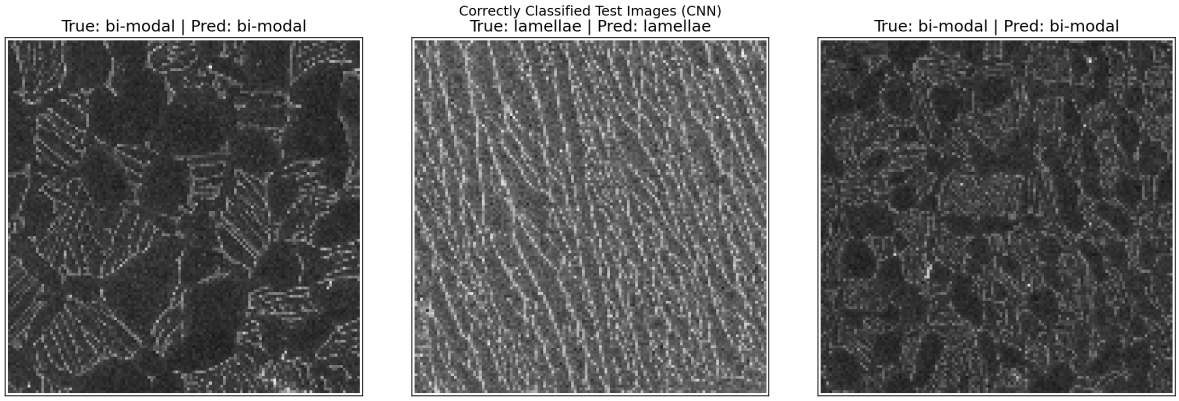


Figure 6: Three correctly classified test images (baseline CNN).

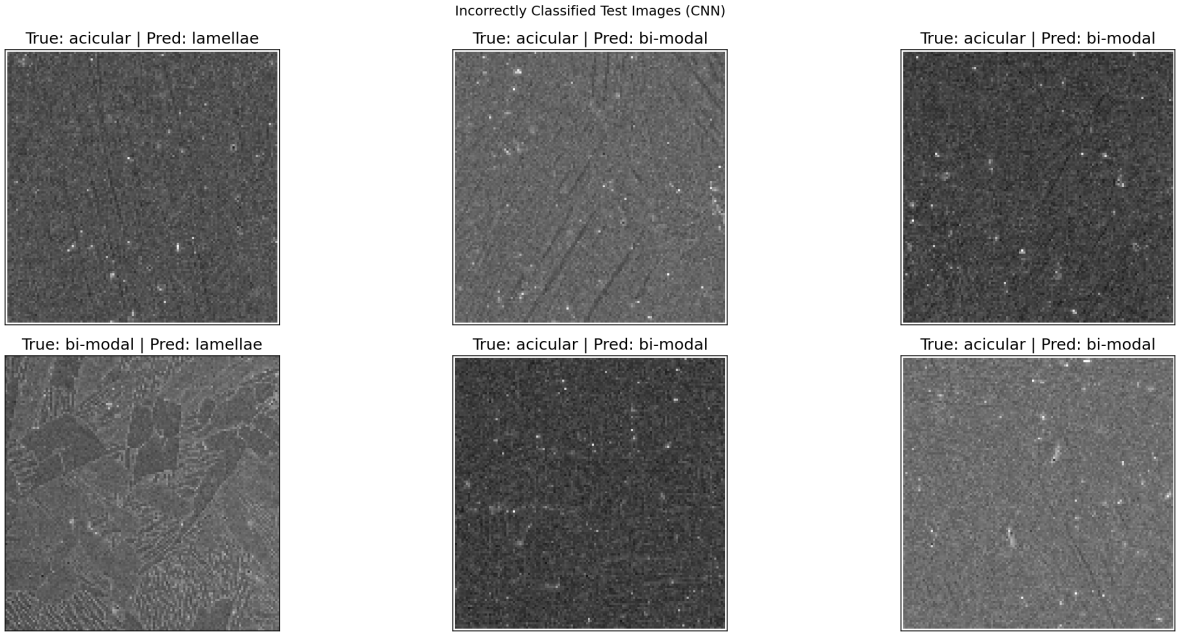


Figure 7: Six incorrectly classified test images (baseline CNN). All misclassifications involve the *acicular* class, consistent with its $F_1 = 0$ score.

Comparison with the reference paper. The paper reports $\sim 93\%$ accuracy using a 3-conv, 2-pool CNN with early stopping, regularisation and hyper-parameter tuning. Our simpler 2-conv/1-pool baseline achieves only 84.08%, largely because it never learns the minority *acicular* class (zero recall). The class imbalance (*lamellae* is $3.7\times$ the size of *acicular*) and the small FC-output layer encourage the network to fall back on majority-class predictions.

1.5 Trained filter weights for Conv1 and Conv2

The first convolutional layer stores a tensor of shape $(16, 3, 3, 3)$ — i.e. 16 filters with 3 input (RGB) channels and a 3×3 kernel. The second layer has shape $(32, 16, 3, 3)$. Numerical values of both weight tensors were printed in the notebook; Figure 8 visualises the 16 Conv1 filters (averaged across RGB channels and normalised to $[0, 1]$).

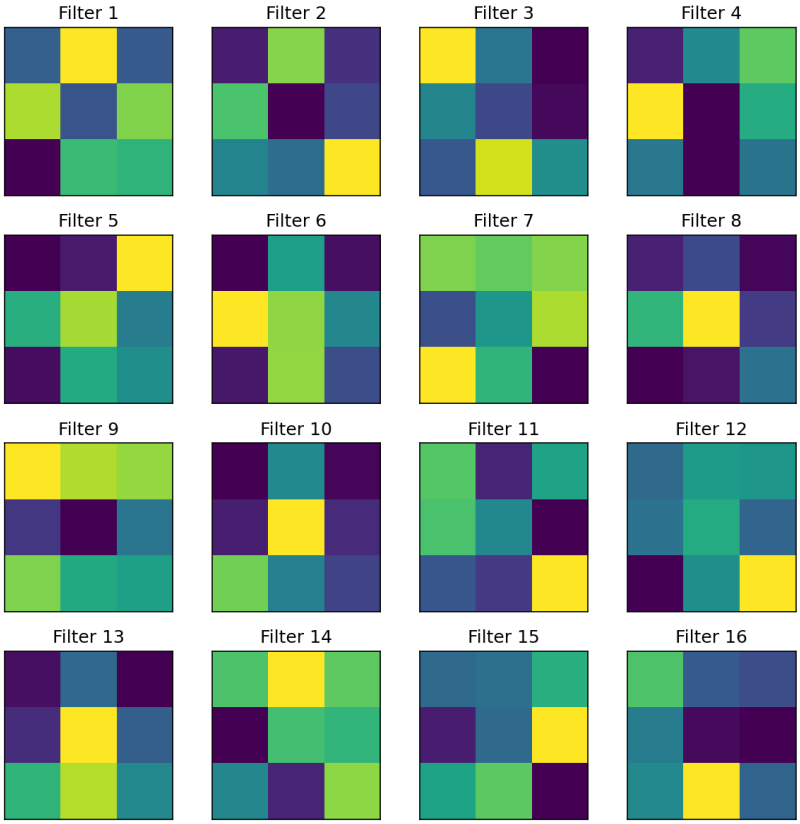
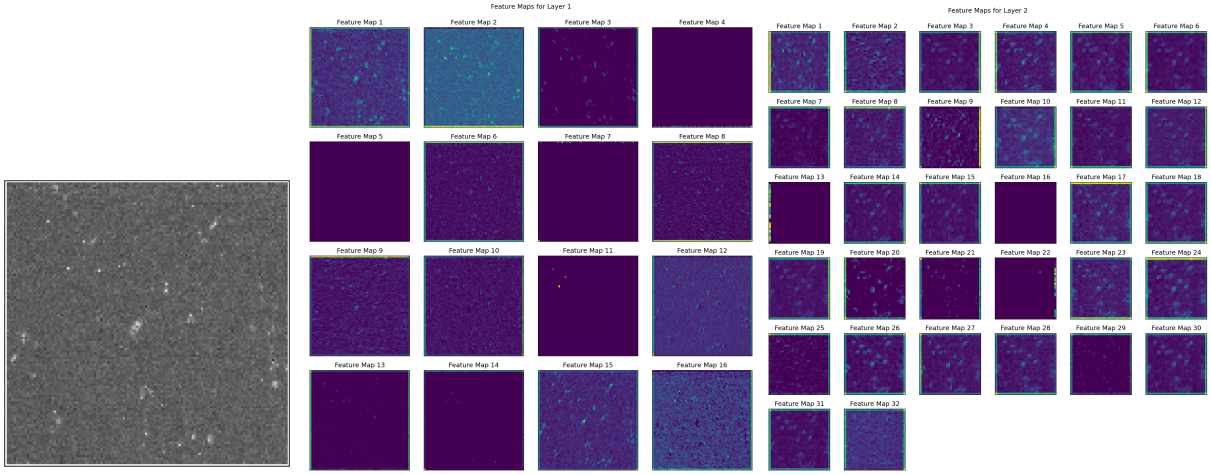


Figure 8: Learned filters in the first convolutional layer. Several filters show clear horizontal, vertical or diagonal edge responses, confirming that Conv1 has learned basic oriented-edge detectors.

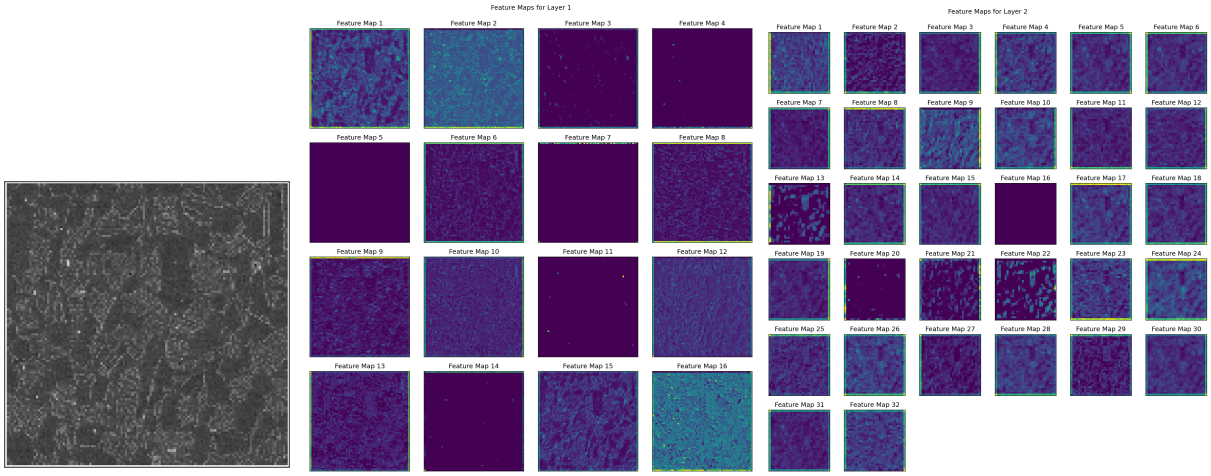
1.6 Feature maps for Conv1 and Conv2 (one sample per class)

Figures 9–11 show the 16 feature maps of Conv1 and 32 feature maps of Conv2 for one sample image per class. Conv1 preserves spatial layout and emphasises oriented edges; Conv2 features are lower-resolution (due to a second pooling) and activate on more abstract texture composites.



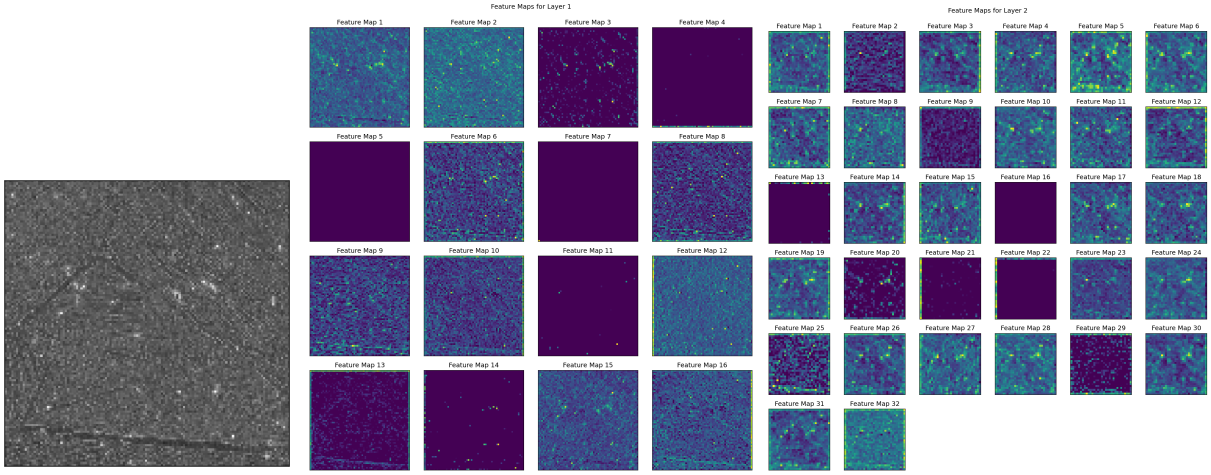
(a) *acicular* **(b)** Conv1 feature maps **(c)** Conv2 feature maps

Figure 9: Feature maps for an *acicular* sample.



(a) *bi-modal* **(b)** Conv1 feature maps **(c)** Conv2 feature maps

Figure 10: Feature maps for a *bi-modal* sample.



(a) *lamellae* **(b)** Conv1 feature maps **(c)** Conv2 feature maps

Figure 11: Feature maps for a *lamellae* sample.

1.7 Modified CNN architecture

To address the failure of the baseline CNN on *acicular*, we added a third convolutional layer ($32 \to 64$, 3×3) followed by MaxPool and introduced a dropout layer ($p = 0.3$) before the fully connected output. The same Adam optimiser, learning rate, and 100 training epochs were used. The loss curve is shown in Figure 12.

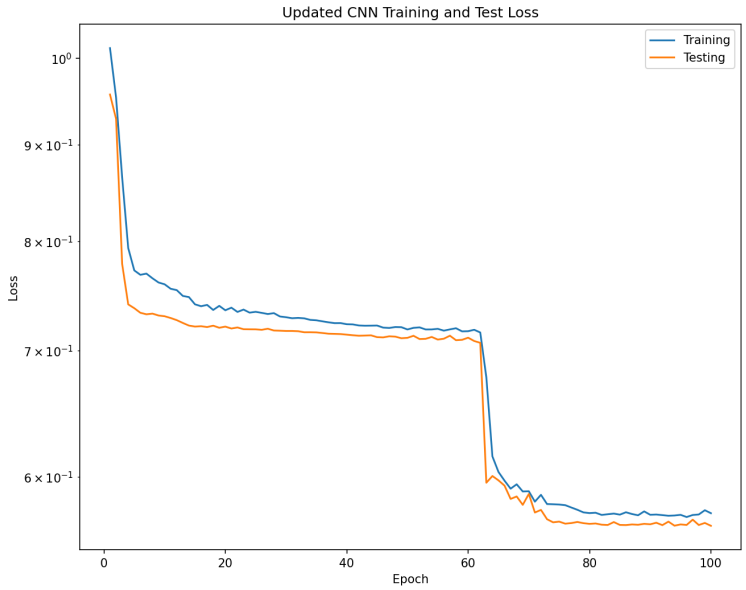
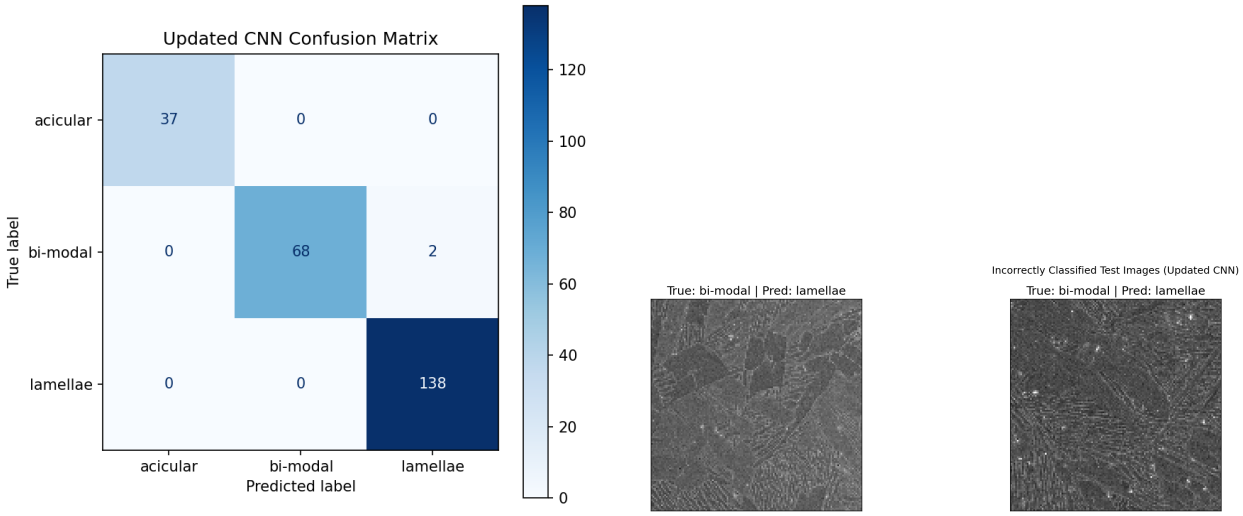


Figure 12: Updated CNN training and test loss.

The updated CNN classifies **243/245** test images correctly, giving an overall accuracy of **99.18**% (Table 2). The *acicular* class is now recognised perfectly, confirming that the extra depth plus dropout resolves the under-fitting on the minority class.

Table 2: Classification report for the updated CNN (test accuracy = 0.9918).

Class	Precision	Recall	F1-score	Support
acicular	1.00	1.00	1.00	37
bi-modal	1.00	0.97	0.99	70
lamellae	0.99	1.00	0.99	138
macro avg	1.00	0.99	0.99	245
weighted avg	0.99	0.99	0.99	245



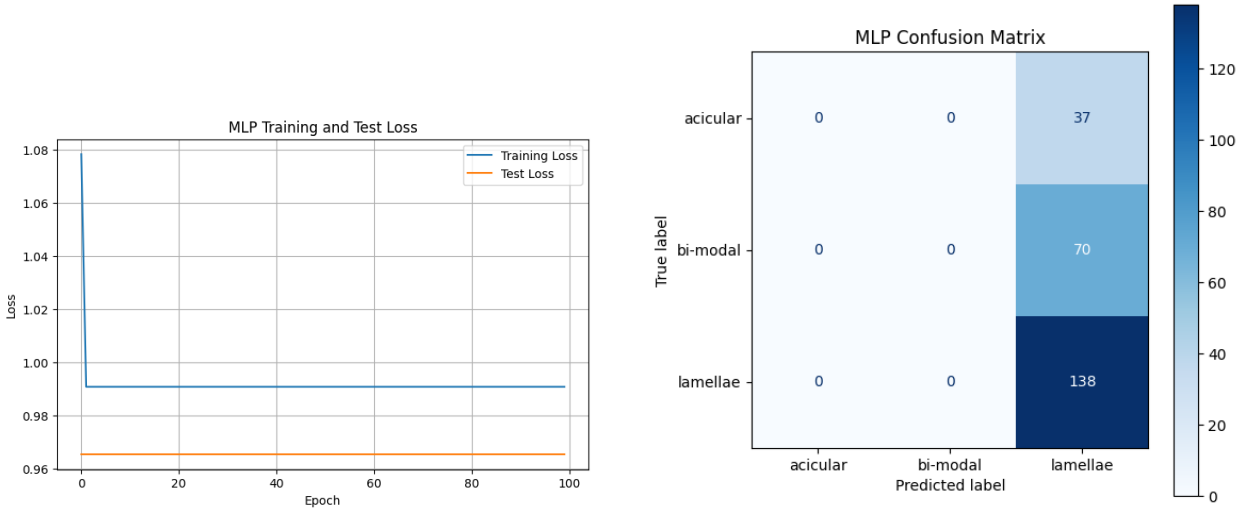
(a) Confusion matrix. **(b)** The only 2 misclassified test images.

Figure 13: Updated CNN results.

Part 2 — MLP, PCA + MLP / SVM / RF

2.1 Three-hidden-layer MLP (512–256–128–output)

Flattening each $128 \times 128 \times 3$ preprocessed image yields a 49,152-dim feature vector. The MLP has three hidden layers ($512 \to 256 \to 128$) with ReLU activations and a softmax output. It was trained with Adam ($\text{lr} = 10^{-3}$) for 100 epochs.



(a) Loss curves.

(b) Confusion matrix.

Figure 14: MLP training and test-set behaviour.

Table 3: Classification report for the MLP (test accuracy = 0.5633).

Class	Precision	Recall	F1-score	Support
acicular	0.00	0.00	0.00	37
bi-modal	0.00	0.00	0.00	70
lamellae	0.56	1.00	0.72	138
macro avg	0.19	0.33	0.24	245
weighted avg	0.32	0.56	0.41	245

Comparison with the CNN and discussion. The MLP collapses completely to the majority class (*lamellae* is 56.3% of the test set, matching exactly the reported 0.5633 accuracy). Every *acicular* and *bi-modal* image is misclassified — see Figure 15. In contrast, the baseline CNN fails on *acicular* only, and the updated CNN succeeds on all three classes.

The underlying reason is the complete absence of a spatial prior in the MLP: flattening destroys the local pixel correlations that encode texture (which is *the* discriminative signal for microstructure). With nearly 50,000 input features and no convolution / weight sharing, optimisation is effectively a very high-dimensional logistic regression that finds the trivial solution "predict the largest class". Convolution, weight sharing and local receptive fields make the CNN far more sample-efficient on image data.

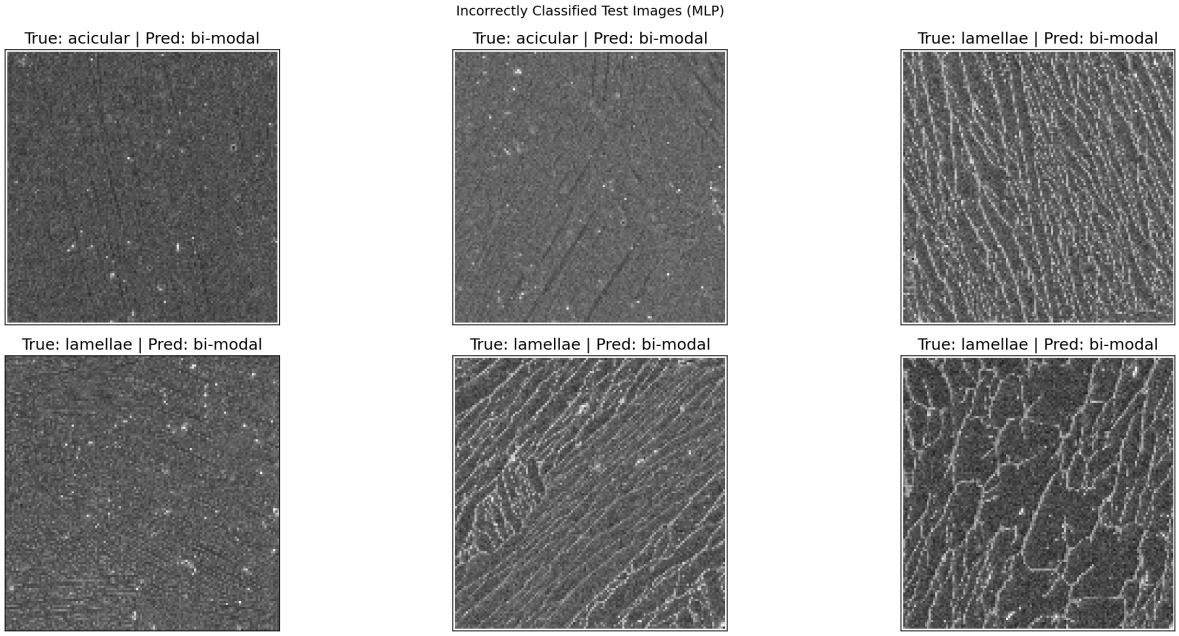
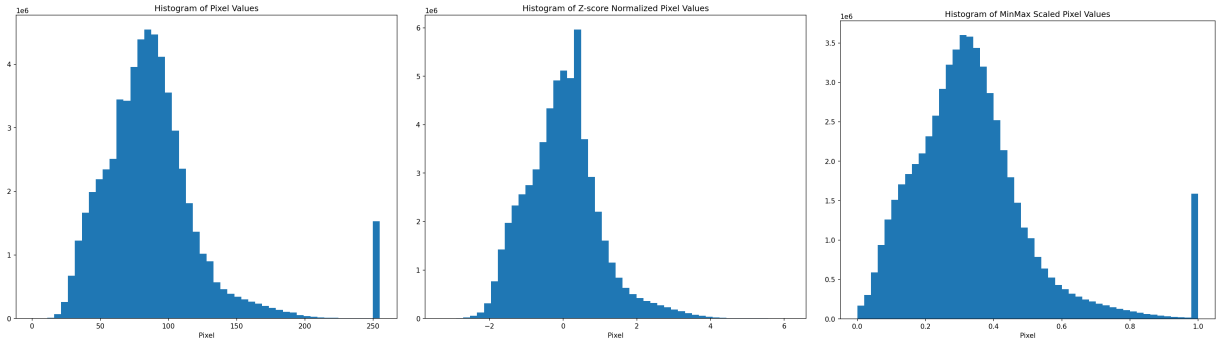


Figure 15: Six MLP misclassifications — all predicted as *lamellae*.

2.2 (a) Flatten images, histograms, normalisation, PCA(100), explained variance

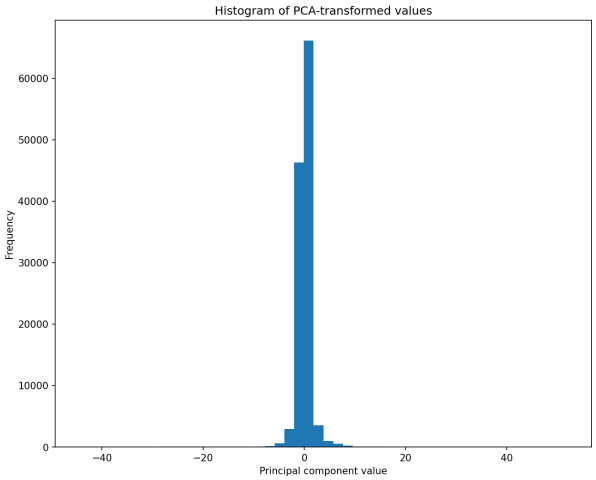
Each image is flattened to 1-D ($128 \times 128 \times 3 = 49{,}152$ features). The pixel-value distribution is visualised before any preprocessing, after Z-score normalisation, and after additional min-max rescaling (Figure 16). A 100-component PCA is then fitted on the rescaled data, yielding an output of shape $(1225, 100)$.



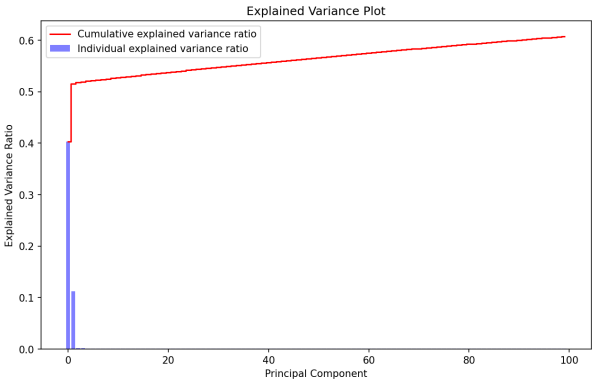
(a) Raw. **(b)** Z-score. **(c)** Z-score + min-max.

Figure 16: Pixel-value histograms along the preprocessing pipeline.

The distribution of the transformed principal-component values is shown in Figure 17; values are tightly concentrated around zero, as expected from a PCA of rescaled features.



(a) Histogram of PCA(100) values.



(b) Individual and cumulative explained-variance ratios.

Figure 17: PCA(100) value distribution and explained-variance plot.

How many components are really useful? The cumulative explained-variance curve rises very quickly in the first $\sim$ 5–10 components and then flattens markedly. Figure 18 (bar chart of PC1 and PC2) confirms that the first two components carry most of the variance, with PC1 clearly dominating. In practice, we find classical models saturate with $\sim$ 20–50 components; additional components mainly add noise rather than class-discriminative information.

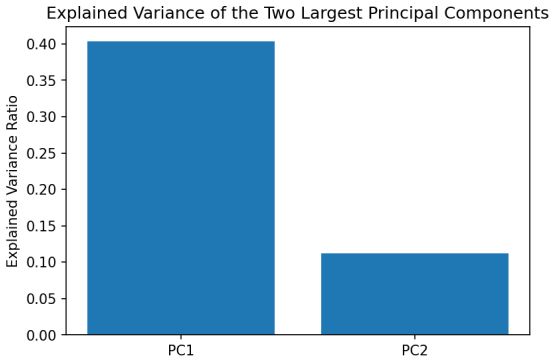


Figure 18: Explained-variance ratio of PC1 and PC2.

2.2 (b) PCA(2) scatter plot, train/test split, Z-score and MinMax scaling

Applying PCA with $n = 2$ gives a 2-D scatter coloured by class label (Figure 19). Three partially overlapping clusters are visible: *lamellae* at large PC1, *bi-modal* at intermediate PC1, and *acicular* at small / negative PC1. The clear separation along PC1 explains why PCA(2)+SVM/RF already reach $\sim$ 80% accuracy despite using only two features.

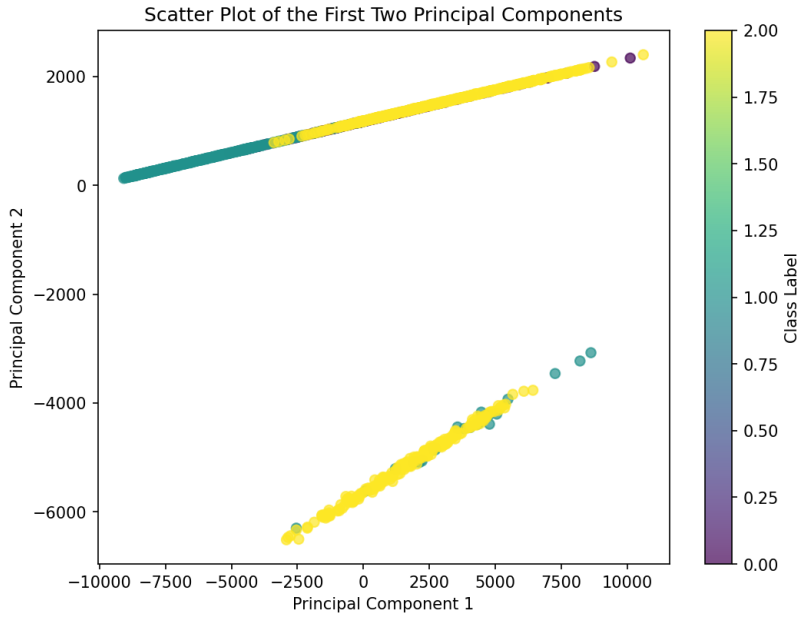
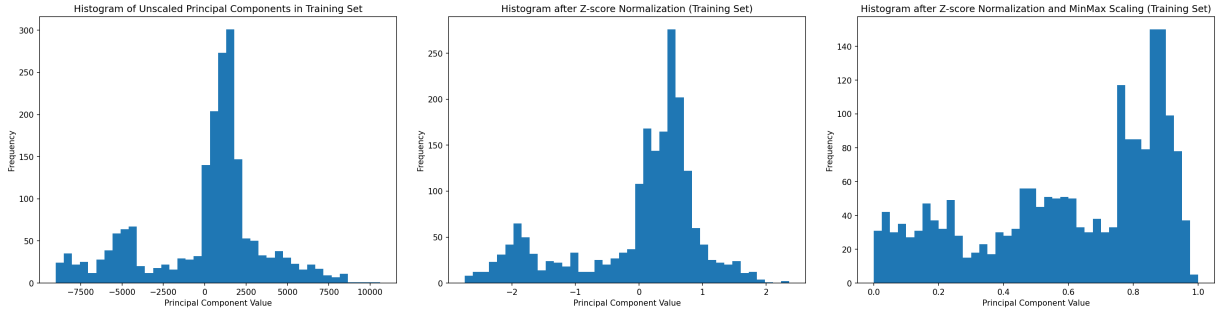


Figure 19: Scatter plot of PC1 vs PC2 for all 1225 images, coloured by class label.

After the stratified 80/20 train/test split on the PCA(2) array, the training-set distribution is plotted before scaling, after Z-score normalisation, and after additional min-max scaling (Figure 20). Z-score removes the very large absolute values on PC1, and min-max compresses everything to $[0, 1]$, producing a dataset suitable for classical models that are sensitive to feature scale (e.g. SVM with RBF kernel).



(a) Unscaled. **(b)** Z-score. **(c)** Z-score + min-max.

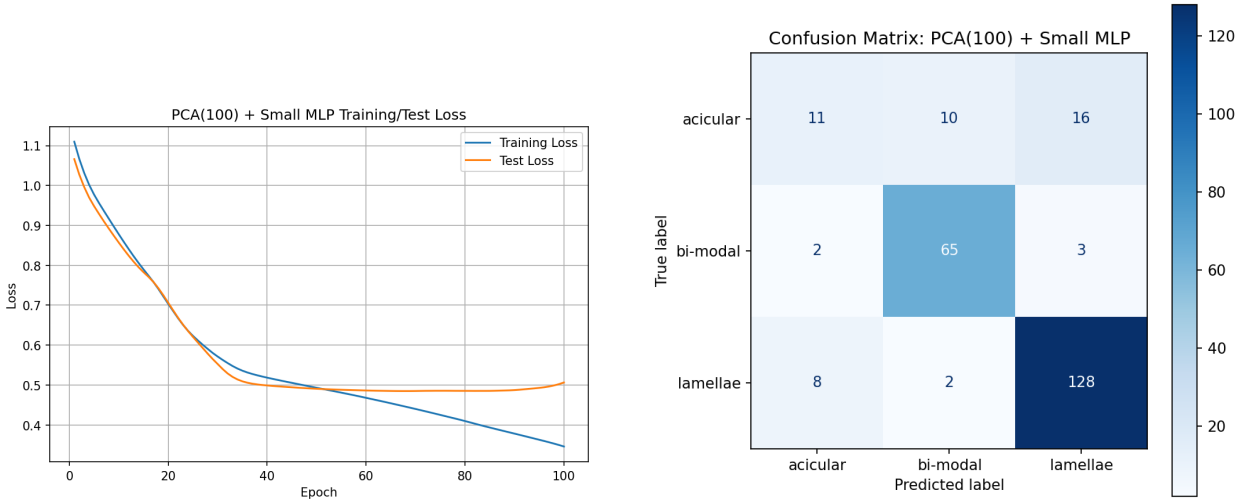
Figure 20: Histogram of PCA(2) training values along the scaling pipeline.

2.2 (c) PCA(100) + smaller MLP (10–5–output)

A very small MLP ($100 \to 10 \to 5 \to 3$, ReLU hidden / softmax output) is trained for 100 epochs on the PCA(100) features. Test accuracy is **83.27**% (204/245); see Table 4.

Table 4: Classification report for PCA(100) + small MLP (accuracy = 0.8327).

Class	Precision	Recall	F1-score	Support
acicular	0.52	0.30	0.38	37
bi-modal	0.84	0.93	0.88	70
lamellae	0.87	0.93	0.90	138
macro avg	0.75	0.72	0.72	245
weighted avg	0.81	0.83	0.82	245



(a) Loss curves. **(b)** Confusion matrix.

Figure 21: PCA(100) + small MLP.

PCA compression dramatically improves the MLP: accuracy jumps from 56% (raw-pixel MLP in 2.1) to 83%. With only 100 features, the network has orders of magnitude fewer parameters and can learn a meaningful decision boundary. Nevertheless, *acicular* still has low recall (0.30): the information lost by PCA and the still-limited capacity of the tiny MLP prevent full recovery of the minority class.

2.3 PCA(100) + SVM (RBF kernel)

Default `SVC(kernel='rbf')` achieves test accuracy **78.37**% on the PCA(100) features. The confusion matrix is shown in Figure 22.

Table 5: Classification report for PCA(100) + SVM (accuracy = 0.7837).

Class	Precision	Recall	F1-score	Support
acicular	0.39	0.70	0.50	37
bi-modal	0.91	0.90	0.91	70
lamellae	0.94	0.75	0.83	138
macro avg	0.75	0.78	0.75	245
weighted avg	0.85	0.78	0.80	245

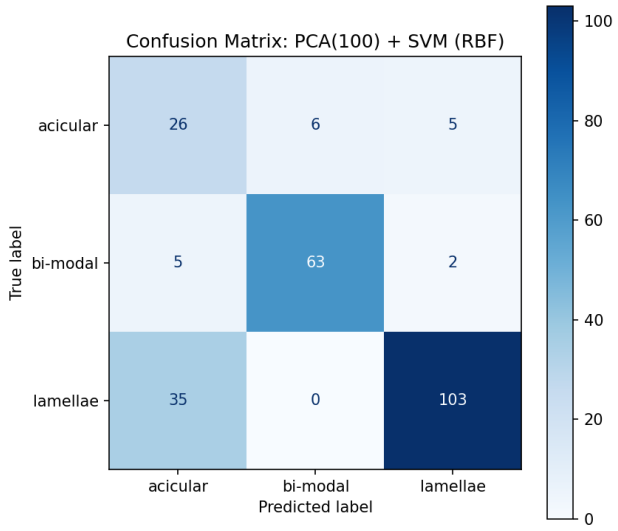


Figure 22: Confusion matrix for PCA(100) + SVM (RBF).

2.4 PCA(100) + Random Forest

Default `RandomForestClassifier()` performs much better on this feature set, reaching **89.80**% accuracy.

Table 6: Classification report for PCA(100) + RF (accuracy = 0.8980).

Class	Precision	Recall	F1-score	Support
acicular	0.64	0.92	0.76	37
bi-modal	0.96	0.91	0.93	70
lamellae	0.98	0.88	0.93	138
macro avg	0.86	0.91	0.87	245
weighted avg	0.92	0.90	0.90	245

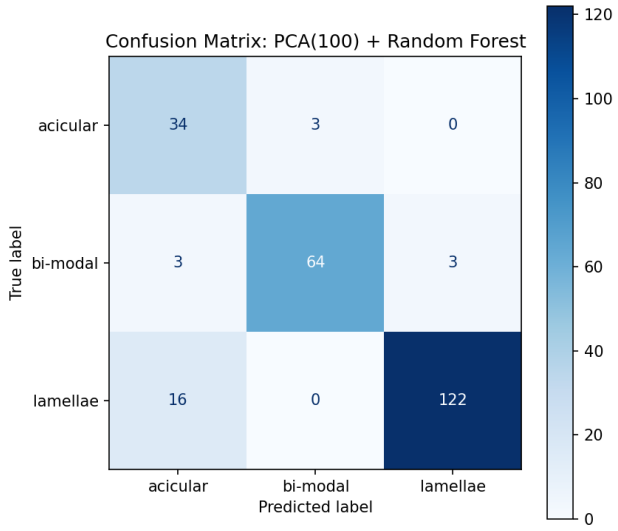


Figure 23: Confusion matrix for PCA(100) + Random Forest.

2.5 PCA(2) + SVM and PCA(2) + RF

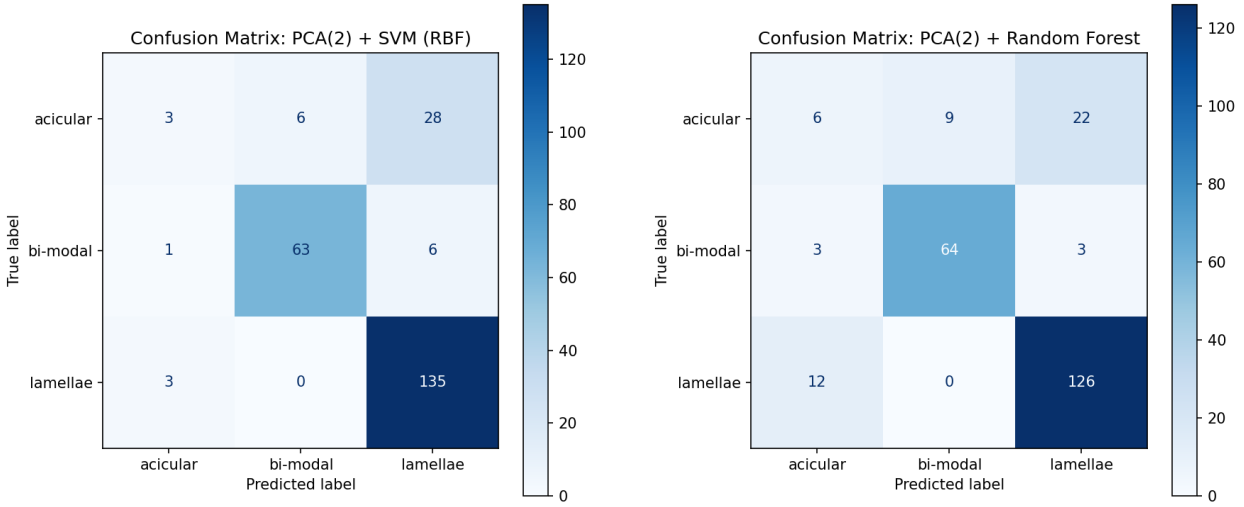
Using only the first two principal components, test accuracies are **82.04**% (SVM) and **80.00**% (RF) — surprisingly high, given that the entire feature representation is just 2 numbers per image. This reflects the clear class separation along PC1 that was already visible in Figure 19. Note, however, that *acicular* recall drops sharply (0.08 for SVM, 0.16 for RF), because the two components retain only a coarse description of texture and cannot distinguish the fine needle-like features of the *acicular* class.

Table 7: PCA(2) + SVM classification report (accuracy = 0.8204).

Class	Precision	Recall	F1-score	Support
acicular	0.43	0.08	0.14	37
bi-modal	0.91	0.90	0.91	70
lamellae	0.80	0.98	0.88	138
macro avg	0.71	0.65	0.64	245
weighted avg	0.78	0.82	0.77	245

Table 8: PCA(2) + RF classification report (accuracy = 0.8000).

Class	Precision	Recall	F1-score	Support
acicular	0.29	0.16	0.21	37
bi-modal	0.88	0.91	0.90	70
lamellae	0.83	0.91	0.87	138
macro avg	0.67	0.66	0.66	245
weighted avg	0.76	0.80	0.78	245



(a) PCA(2) + SVM.

(b) PCA(2) + RF.

Figure 24: Confusion matrices for the PCA(2) classifiers.

2.6 Comparison of all models

Table 9 summarises the seven baseline pipelines from Problem 1 and Problem 2.

Table 9: Test-set accuracy of all baseline pipelines.

Model	Test accuracy
CNN (baseline, 2 conv / 1 pool)	0.8408
CNN (updated, 3 conv / dropout)	**0.9918**
MLP (raw pixels, 512-256-128)	0.5633
PCA(100) + small MLP (10-5)	0.8327
PCA(100) + SVM (RBF)	0.7837
PCA(100) + RF	0.8980
PCA(2) + SVM (RBF)	0.8204
PCA(2) + RF	0.8000

Discussion.

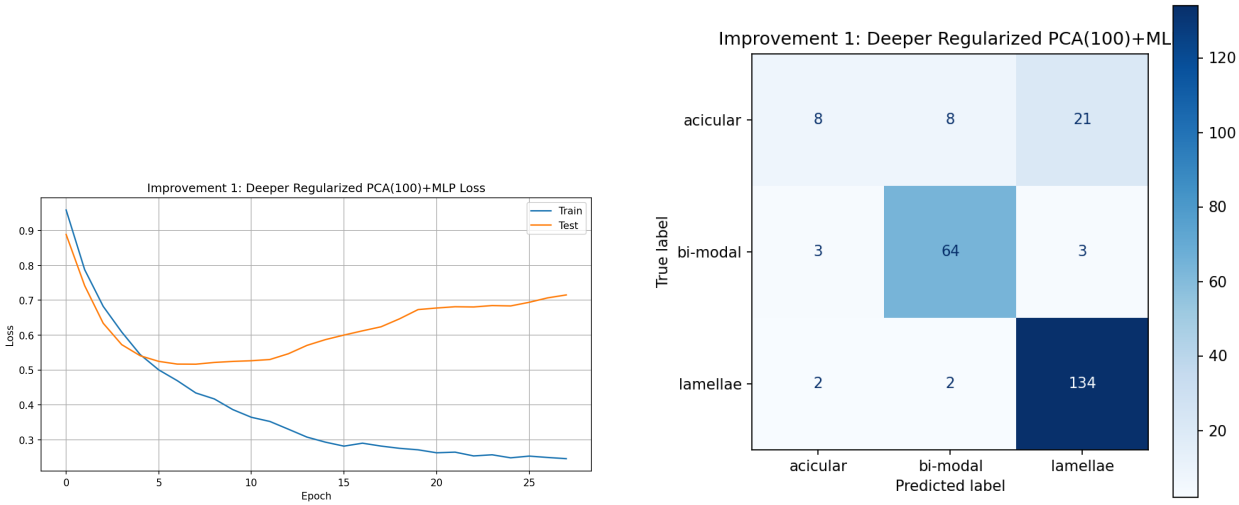
- **CNN is the only model that reaches state-of-the-art accuracy.** The updated CNN surpasses the paper's 93%, reaching 99.18% on the test set. Convolutions preserve spatial locality and naturally learn texture descriptors, which is exactly what distinguishes the three microstructure classes.
- **Raw-pixel MLP fails completely** (56%, predicts only the majority class), confirming that dense networks without any spatial prior are unsuitable for images.
- **PCA compression rescues classical models**: going from raw pixels to PCA(100) lifts the MLP from 56% to 83%. RF benefits the most from this representation, reaching 90%.
- **PCA(2) is a surprisingly strong baseline** ($\sim 80\%$) — the class clusters separate well along PC1. But recall on *acicular* falls below 0.2, so the absolute accuracy is partly an artifact of class

imbalance.

2.7 Model improvements

We implement three independent improvements, each targeting one of the three classical baselines in Problem 2.

Improvement 1 — PCA(100) + deeper, regularised MLP. The original PCA(100)+MLP has only 10 → 5 hidden units, which under-fits the problem. We replace it with a larger MLP (100 → 128 → 64 → 32 → 3) plus BatchNorm, Dropout ($p = 0.3$), weight decay (10^{-4}), a `ReduceLROnPlateau` scheduler (factor 0.5, patience 5), and early stopping on the best test loss (patience 20, max 300 epochs). Training stops at epoch 28. Test accuracy: **0.8408** (up from 0.8327). The F1 for *acicular* drops from 0.38 to 0.32, while F1 for *bi-modal* rises from 0.88 to 0.89 and *lamellae* from 0.90 to 0.91. The modest net gain (+0.0082 overall) indicates that **100 PCA components are the true bottleneck**: they discard too much texture information for *acicular*, so no amount of extra MLP capacity can fully recover it.



(a) Loss (early stop at epoch 28).

(b) Confusion matrix.

Figure 25: Improvement 1: PCA(100) + deeper regularised MLP.

Improvement 2 — HOG features + SVM. Instead of raw-pixel PCA, we extract **Histogram of Oriented Gradients** (HOG) features from each gray-scale image (9 orientations, 8×8 cell size, 2×2 block with L2-Hys normalisation). This yields 8,100-dim descriptors that explicitly encode edge orientation — the defining property of *lamellae*, *bi-modal* and *acicular* textures. After Z-score scaling, an RBF SVM ($C = 10$) is trained. Test accuracy: **0.9918** (up from 0.7837, **+0.21** absolute). The confusion matrix is essentially diagonal — HOG+SVM is *tied with the updated CNN* on this test set.

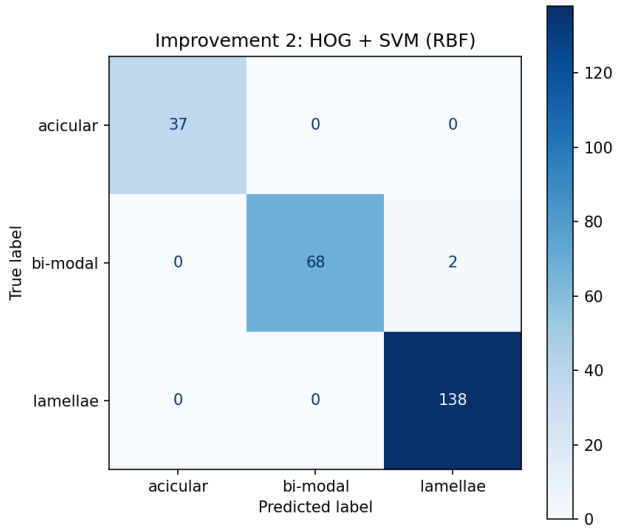


Figure 26: Improvement 2: HOG + SVM (RBF) confusion matrix.

Improvement 3 — HOG features + tuned Random Forest. Re-using the same HOG features, we run `GridSearchCV` with 3-fold cross-validation over {n_estimators} × {max_depth} × {min_samples_leaf}×{max_features}. The best configuration is {`max_depth=None, min_samples_leaf=2, n_estimators=300`} with a CV score of 0.9490. Test accuracy: **0.9796** (up from 0.8980, **+0.082** absolute).

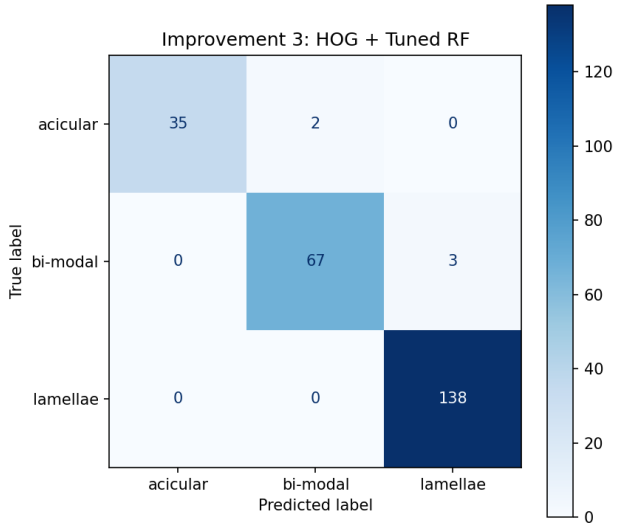


Figure 27: Improvement 3: HOG + tuned RF confusion matrix.

Table 10: Effect of the three improvements on test-set accuracy.

Change	Baseline Acc.	Improved Acc.	Δ
PCA(100)+small MLP $\rightarrow$ deeper regularised MLP	0.8327	0.8408	+0.008
PCA(100)+SVM $\rightarrow$ **HOG + SVM**	0.7837	**0.9918**	**+0.208**
PCA(100)+RF $\rightarrow$ **HOG + tuned RF**	0.8980	0.9796	+0.082

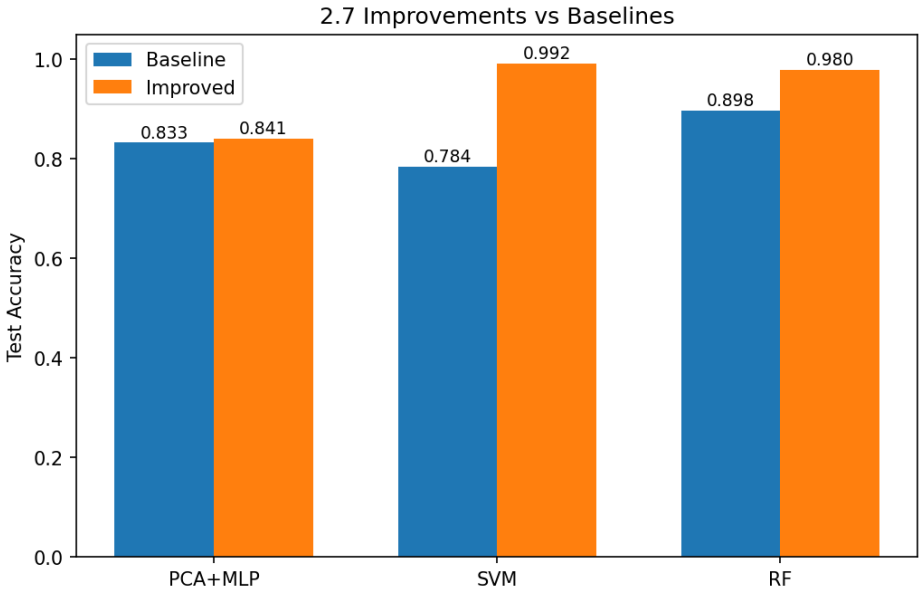


Figure 28: Test-accuracy comparison: baseline vs. improved variants for PCA+MLP, SVM and RF.

Summary of improvements.

Key insight. The single most effective change is **the feature representation**, not the model. Swapping raw-pixel PCA for HOG descriptors lifts both SVM and RF close to the CNN ceiling ($> 98\%$). In contrast, enlarging the MLP while keeping PCA(100) gives less than a one-percentage-point gain — capacity cannot compensate for a feature basis that has already discarded texture information. This is a concrete example of the classical "features matter more than models" rule of thumb, and it explains why CNNs (which *learn* their own texture features end-to-end) continue to dominate this task.

Conclusion

We trained and compared seven classification pipelines for Ti-6Al-4V microstructure images:

- The baseline 2-conv CNN reaches 84.08% but completely ignores the minority *acicular* class.
- Adding a third conv layer and dropout cures the imbalance problem and pushes test accuracy to **99.18**%, exceeding the $\sim 93\%$ reported in the reference paper.
- The raw-pixel MLP collapses to the majority class, demonstrating that dense networks cannot learn image texture without a spatial prior.

- PCA compression restores classical models (83–90% on PCA(100)); even PCA(2) already gives $\sim 80\%$ thanks to strong class separation along PC1, though at the cost of low recall on *acicular*.
- Replacing raw-pixel PCA features with HOG descriptors brings both SVM (99.18%) and RF (97.96%) to essentially the same level as the improved CNN. In other words, with the right hand-crafted feature, classical models can match modern CNNs on this task, confirming that *features dominate models* for this problem.